COURSE SLIDES · WSQ

ISO 14064-1:2018 Greenhouse Gas (GHG) Emissions and Removals



WSQ Course Code: TGS-2025055502

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v2.0 · 20 August 2026

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved.

COURSE ADMINISTRATION

Welcome and Course Orientation

© 2026 Tertiary Infotech Academy Pte Ltd

Digital Attendance (Mandatory)

1

AM, PM and assessment

Complete every required SSG digital-attendance session.

2

Scan

Use the SSG QR code shown by the trainer or administrator.

3

Confirm

Wait for the successful submission acknowledgement.

4

Eligibility

At least 75% attendance is required for assessment and funding.

© 2026 Tertiary Infotech Academy Pte Ltd

About the Trainer



Your Trainer

Complete this profile before class

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

© 2026 Tertiary Infotech Academy Pte Ltd

NAMED TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE

Principal Trainer and course developer

EXPERTISE

Sustainability systems, organisational GHG accounting and business improvement

DELIVERS

WSQ professional and technical programmes

FOUNDER

Tertiary Infotech / Tertiary Courses

© 2026 Tertiary Infotech Academy Pte Ltd

LEARNER INTRODUCTION

Let's Know Each Other

1

You

Name, organisation and role.

2

Your footprint

One emission source or climate issue relevant to your work.

3

Your challenge

A recurring GHG data, reporting or reduction problem.

4

Your outcome

What successful learning looks like after two days.

© 2026 Tertiary Infotech Academy Pte Ltd

HOUSEKEEPING

Ground Rules

1. **Be present**
   Return from breaks on time and complete attendance.

2. **Participate**
   Ask, challenge and contribute evidence.

3. **Respect**
   One conversation at a time; disagree professionally.

4. **Protect information**
   Use fictional or sanitised workplace data.

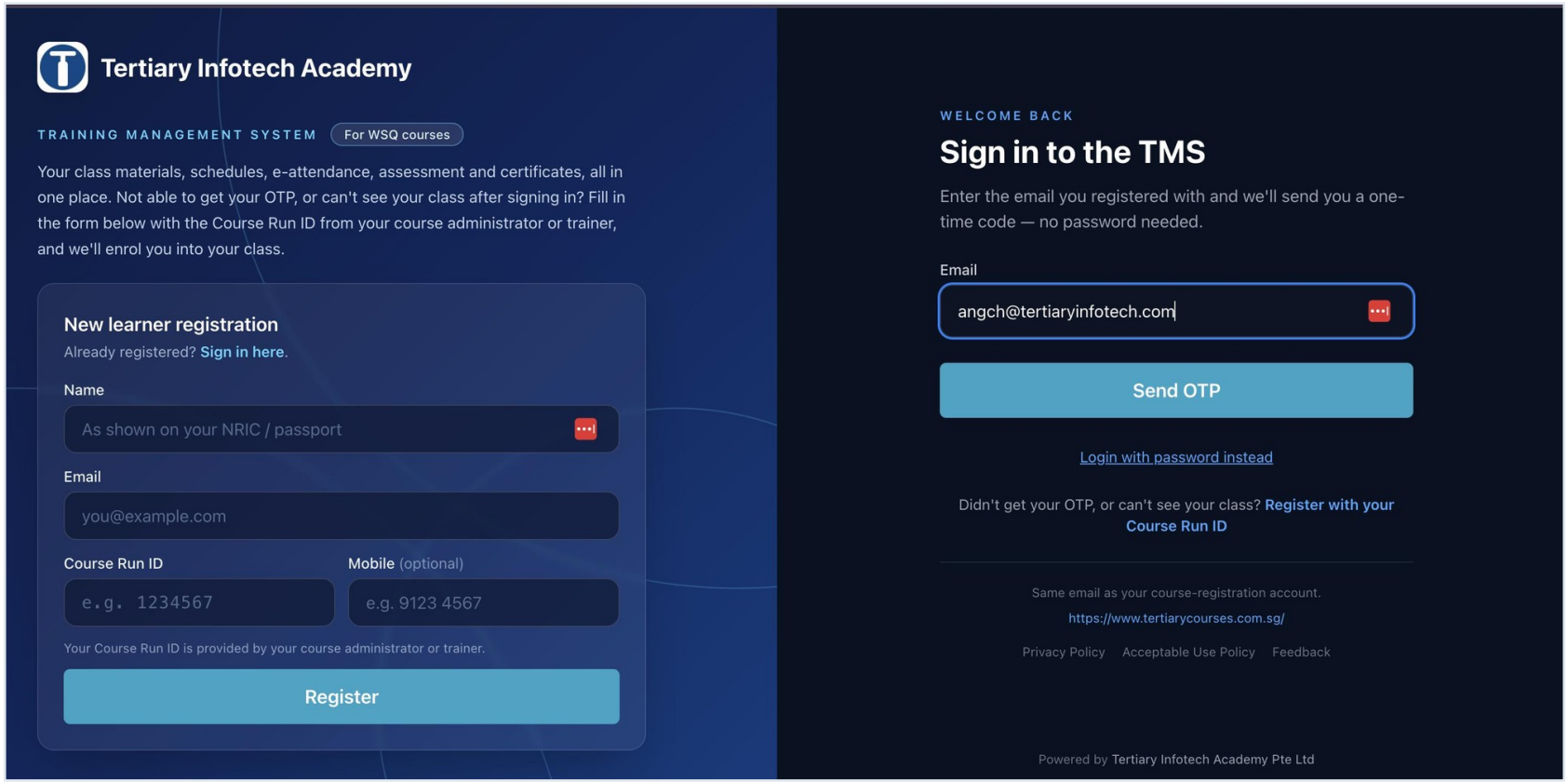
5. **Stay practical**
   Link every idea to a process or decision.

6. **Own the result**
   Verify your activity output before submission.

© 2026 Tertiary Infotech Academy Pte Ltd

COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

© 2026 Tertiary Infotech Academy Pte Ltd

SCHEDULE

Lesson Plan — Two Days

Day 1

- Day 1 · ISO 14064-1 inventory design, boundaries, quantification, reporting and risk controls

– 9:00-10:30 · administration, GHG science and ISO family

– 10:40-13:00 · boundaries and Activities 1-2

– 14:00-15:20 · quantification and Activity 3

– 15:30-18:00 · review, governance and Activities 4-5

Day 2

- Day 2 · Value-chain reduction, renewable energy, offsets and integrated carbon strategy

– 9:00-10:30 · hotspots, lifecycle and Activity 6

– 10:40-13:00 · supplier roadmap and renewable energy

– 14:00-15:20 · offset integrity and Activity 9

– 15:30-16:00 · Activity 10 and wrap-up

– 16:00-18:00 · WA then Case Study

© 2026 Tertiary Infotech Academy Pte Ltd

Learning Outcomes

1

LO1

Innovate organisational GHG accounting systems using current climate science, ISO 14064-1 principles and broader sustainability initiatives.

2

LO2

Develop risk-informed GHG management plans that meet relevant reporting and regulatory requirements and support mitigation decisions.

3

LO3

Develop operational emissions-reduction strategies across product lifecycles, value chains, supply chains and renewable-energy options.

4

LO4

Generate economically viable carbon-offset and carbon-credit strategies aligned with organisational targets and residual-emissions needs.

© 2026 Tertiary Infotech Academy Pte Ltd

Knowledge Alignment · K1-K5

1

K1

Evolving research on GHG dynamics and innovation in GHG accounting

2

K2

Integration of GHG accounting with sustainability and climate initiatives

3

K3

GHG reduction across product lifecycles, value chains and supply chains

4

K4

Renewable-energy technologies, economics, sustainability impacts and regulation

5

K5

Best-practice approaches for GHG accounting and reporting

© 2026 Tertiary Infotech Academy Pte Ltd

Knowledge Alignment · K6-K9

1

K6

Relevant regulations, programmes and reporting guidelines

2

K7

Frameworks for evaluating, mitigating and monitoring GHG-related risks

3

K8

Approaches and economics of carbon-offset projects

4

K9

Sustainability frameworks integrated into GHG-reduction strategy

© 2026 Tertiary Infotech Academy Pte Ltd

Ability Alignment · A1-A4

1

A1

Develop reduction strategies across product lifecycles, value chains and supply chains

2

A2

Generate the economic costs and benefits of offset projects to determine suitability

3

A3

Innovate organisational GHG accounting using emerging science and accounting practice

4

A4

Develop GHG management frameworks, plans and monitoring initiatives

© 2026 Tertiary Infotech Academy Pte Ltd

Ability Alignment · A5-A8

1

A5

Develop a carbon strategy for project selection and credit optimisation

2

A6

Influence economically viable renewable-energy strategies aligned with sustainability goals

3

A7

Develop GHG accounting and reporting guidelines integrated with sustainability initiatives

4

A8

Define assessments and mitigation strategies for GHG-related risks

© 2026 Tertiary Infotech Academy Pte Ltd

ASSESSMENT BRIEFING

Briefing for Assessment

1

Individual

No discussion; answer in your own words.

2

Open book

Use slides, Learner Guide and approved activity material only.

3

Secure

Place phones and other materials under the table; no recording.

4

Submission

Complete the provided document and upload to the LMS.

5

Competency

Every assessed K and A criterion must be Competent.

6

Rights

Re-assessment and appeal routes are available.

© 2026 Tertiary Infotech Academy Pte Ltd

FINAL ASSESSMENT

Assessment and Funding

Written Assessment

- 9 open-ended questions
- K1-K9 · 1 hour
- Open book

Case Study

- One coherent scenario
- A1-A8 · 1 hour
- Open book

Competency

- C in both instruments
- At least 75% attendance
- Funding criteria apply

© 2026 Tertiary Infotech Academy Pte Ltd

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM digital attendance
2. Assessment digital attendance
3. Sit WA then Case Study
4. Submit answers on the LMS
5. Sign Assessment Summary Record

© 2026 Tertiary Infotech Academy Pte Ltd

SELF-CONTAINED ACTIVITY PACKS

Access the Course Activities

Option A · Clone

- Open the repository link
- Copy the HTTPS clone URL
- Open the activities folder

Option B · ZIP

- GitHub · Code
- Download ZIP
- Extract and open activities/

Open the ISO 14064-1 activity repository

Each folder

- Detailed README.md
- Working XLSX
- Printable PDF/checklist

© 2026 Tertiary Infotech Academy Pte Ltd

ACTIVITY EVIDENCE WORKFLOW

Use Evidence Before Claims

STEP 1 OF 1

1

Define the decision, document the boundary, calculate transparently, review the evidence and retain the approved output.

© 2026 Tertiary Infotech Academy Pte Ltd

https://lms-tms.tertiaryinfotech.com/

Course Portal

- Download the learner slides and Learner Guide.
- Open the complete Activities folder and working files.
- Complete and submit assessments only during the scheduled window.

© 2026 Tertiary Infotech Academy Pte Ltd

01

TOPIC 01

Foundations of Organisational GHG Accounting Systems

GHG science · ISO 14064 family · accounting principles · targets · sustainability integration

© 2026 Tertiary Infotech Academy Pte Ltd

K1 · K2 · K5 · A3 · A7 · LO1

Topic 01 · Core Model

1

Relevance

Select sources, sinks, reservoirs and methods that serve intended users and decisions.

2

Completeness

Include all significant emissions and removals within justified boundaries.

3

Consistency

Apply methods that support meaningful comparison over time and disclose changes.

4

Accuracy and transparency

Reduce bias and uncertainty, and document assumptions, evidence and limitations.

© 2026 Tertiary Infotech Academy Pte Ltd

K1 · K2 · K5 · A3 · A7 · LO1

GHG science for organisational decisions

1

Radiative forcing

GHGs absorb and re-emit infrared energy, changing the Earth's energy balance.

2

Long-lived stocks

Carbon dioxide accumulates; current emissions add to a persistent atmospheric stock.

3

Different gases

Methane, nitrous oxide and fluorinated gases differ in lifetime and warming effect.

4

CO2e

A common unit converts gas mass using a declared global-warming-potential basis.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Inventory all relevant gases
- State the GWP source and assessment basis
- Use science to improve priorities
- Separate physical fact from target choice

Avoid the trap

- CO2e is a comparison convention
- A lower annual total does not erase historic stock
- GWP values can change by assessment
- Avoid false precision beyond the data

© 2026 Tertiary Infotech Academy Pte Ltd

K1 · K2 · K5 · A3 · A7 · LO1

ISO 14064 family and assurance architecture

1

Part 1

Organisation-level principles and requirements for inventory design, quantification and reporting.

2

Part 2

Project-level quantification, monitoring and reporting of emission reductions or removal enhancements.

3

Part 3

Principles and requirements for verification and validation of GHG statements.

4

Related frameworks

GHG Protocol, ISO 14067 and programme rules can complement—but not silently replace—the chosen criteria.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Name the inventory criteria
- Define the intended assurance use
- Keep organisation and project claims distinct
- Resolve conflicting programme rules

Avoid the trap

- Certification is not automatic
- A project credit is not an organisational removal
- One framework mapping may not be exact
- Use authorised standards text for formal conformance

© 2026 Tertiary Infotech Academy Pte Ltd

K1 · K2 · K5 · A3 · A7 · LO1

ISO 14064-1 principles in practice

1

Relevance

Information and boundaries must serve intended users and decisions.

2

Completeness

Include significant sources and justify omissions transparently.

3

Consistency

Methods and boundaries support comparison through time.

4

Accuracy and transparency

Reduce bias and uncertainty and disclose assumptions, evidence and limitations.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Turn each principle into a review question
- Use materiality and significance deliberately
- Document changes at the time they occur
- Prefer traceable evidence over unsupported detail

Avoid the trap

- Principles interact and require judgement
- Completeness does not mean perfect data
- Consistency does not freeze weak methods
- Transparency does not excuse avoidable error

© 2026 Tertiary Infotech Academy Pte Ltd

K1 · K2 · K5 · A3 · A7 · LO1

Organisational and reporting boundaries

1

Purpose first

Intended users and decisions determine the inventory design.

2

Consolidation

Equity share, financial control or operational control changes which operations are included.

3

Reporting boundary

Direct and significant indirect emissions are organised into decision-useful categories.

4

Boundary evidence

Ownership, leases, contracts, operating authority and influence support the decision.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Map all entities before excluding any
- Apply one consolidation approach consistently
- Document outsourced activities
- Reassess after acquisitions or divestments

Avoid the trap

- Legal ownership and operational control can differ
- Outsourcing does not make emissions disappear
- Boundary choice can change trends
- Avoid cherry-picking low-emission operations

© 2026 Tertiary Infotech Academy Pte Ltd

K1 · K2 · K5 · A3 · A7 · LO1

Science-aligned targets and target adjustments

1

Target boundary

Specify gases, operations, categories and coverage so ambition can be compared with the inventory.

2

Base year

Choose a representative period and publish the recalculation policy before structural change occurs.

3

Pathway

Connect near-term milestones and initiatives to an explicit trajectory rather than a distant headline alone.

4

Adjustment

Recalculate for material structural or methodology changes; do not rewrite history for ordinary performance variance.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Link target scope to inventory scope
- Define a significance threshold
- Separate recalculation from restatement
- Retain approvals and an adjustment bridge

Avoid the trap

- A science-aligned target is not an inventory requirement
- Growth assumptions must remain visible
- Acquisitions can break comparability
- Offsets do not repair an inconsistent base year

© 2026 Tertiary Infotech Academy Pte Ltd

K1 · K2 · K5 · A3 · A7 · LO1

Quantification, base year and report integrity

1

Activity data

Measured or estimated quantities describe the emission-causing activity.

2

Emission factor

A documented coefficient translates activity into gas mass or CO2e.

3

Base year

A stable comparison point needs recalculation rules for structural and methodological change.

4

Report

Disclose boundaries, methods, exclusions, factors, uncertainty, changes and performance separately.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Retain a calculation trail
- Reconcile to primary records
- Recalculate under a written policy
- Report gross inventory before offsets

Avoid the trap

- Units cause frequent large errors
- Spend data can mask physical change
- Estimated data must be labelled
- Credits must not be netted into gross emissions

© 2026 Tertiary Infotech Academy Pte Ltd

Activities · Topic 1

Activity 1

- Set the GHG Inventory Purpose and Boundaries
- K1 · K2 · K5 · A3 · LO1
- 60 minutes

Activity 2

- Build a Significant GHG Source and Sink Register
- K1 · K5 · A3 · A7 · LO1
- 60 minutes

Activity 3

- Quantify a Base-Year Inventory and Reporting Guide
- K1 · K2 · K5 · A3 · A7 · LO1
- 75 minutes

© 2026 Tertiary Infotech Academy Pte Ltd

K1 · K2 · K5 · A3 · LO1 · 60 minutes

Set the GHG Inventory Purpose and Boundaries

ACTIVITY 1

Learners become the GHG inventory team for EcoForge Asia and turn an incomplete corporate structure into a justified organisational and reporting boundary.

You'll build

An inventory purpose statement, facility map, consolidation decision log and reporting-boundary diagram.

Tools: Activity workbook · EcoForge scenario pack · boundary decision checklist

© 2026 Tertiary Infotech Academy Pte Ltd

Activity 1 · Process

1. Review the fact pattern
2. Define inventory purpose
3. Map entities and operations
4. Select consolidation approach
5. Record boundary decisions

© 2026 Tertiary Infotech Academy Pte Ltd

ACTIVITY 1 · ACCEPTANCE CHECK

Set the GHG Inventory Purpose and Boundaries

✅ Test it

The diagram covers every named entity and operation; the consolidation method is applied consistently; every exclusion and unresolved item has a transparent rationale and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

K1 · K5 · A3 · A7 · LO1 · 60 minutes

Build a Significant GHG Source and Sink Register

ACTIVITY 2

Learners inspect EcoForge operations and build a complete source register using ISO 14064-1 categories, with documented significance criteria.

You'll build

A source/sink register, category map, significance assessment and data-owner matrix.

Tools: Source inventory workbook · process narrative · significance criteria cards

© 2026 Tertiary Infotech Academy Pte Ltd

Activity 2 · Process

1. Map the value chain
2. Classify sources and sinks
3. Identify relevant gases
4. Define significance criteria
5. Score indirect sources

© 2026 Tertiary Infotech Academy Pte Ltd

ACTIVITY 2 · ACCEPTANCE CHECK

Build a Significant GHG Source and Sink Register

✅ Test it

The register contains all scenario sources, uses the six-category logic consistently, treats removals separately from offsets and gives evidence-based significance decisions.

© 2026 Tertiary Infotech Academy Pte Ltd

K1 · K2 · K5 · A3 · A7 · LO1 · 75 minutes

Quantify a Base-Year Inventory and Reporting Guide

ACTIVITY 3

Learners convert EcoForge activity data into CO2e, investigate anomalies and draft the accounting rules that make the inventory repeatable.

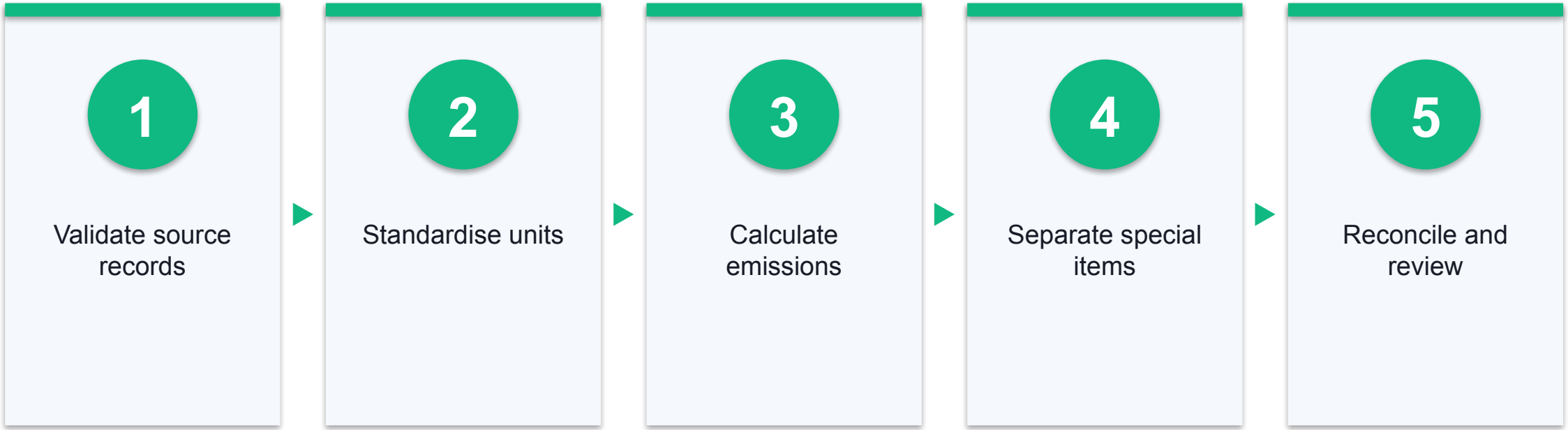
You'll build

A quantified 2025 inventory, calculation trail, uncertainty register, base-year policy and reporting-guideline extract.

Tools: GHG calculation workbook · factor register · sample invoices and meter records

© 2026 Tertiary Infotech Academy Pte Ltd

Activity 3 · Process



© 2026 Tertiary Infotech Academy Pte Ltd

ACTIVITY 3 · ACCEPTANCE CHECK

Quantify a Base-Year Inventory and Reporting Guide

✅ Test it

Every result traces to source data, units, factor and GWP basis; totals reconcile; uncertainty and recalculation triggers are explicit; no offset is netted into the inventory.

© 2026 Tertiary Infotech Academy Pte Ltd

K1 · K2 · K5 · A3 · A7 · LO1 · TOPIC CHECK

Recap · Foundations of Organisational GHG Accounting Systems

1

Explain

Select sources, sinks, reservoirs and methods that serve intended users and decisions.

2

Apply

Define intended users, reporting boundaries and consolidation choices for a credible organisational GHG inventory.

3

Demonstrate

Classify direct and indirect GHG sources, sinks and reservoirs and assess their significance for reporting.

4

Evidence

Retain the completed activity artefacts and review feedback.

© 2026 Tertiary Infotech Academy Pte Ltd

02

TOPIC 02

Regulatory-Compliant and Risk-Responsive GHG Management

Reporting requirements · inventory review · governance · climate risk · control and conformance

© 2026 Tertiary Infotech Academy Pte Ltd

K6 · K7 · A4 · A8 · LO2

Topic 02 · Core Model

1

Governance

Assign authority, competence, control ownership and escalation for inventory quality.

2

Compliance

Maintain a register of applicable laws, schemes, contracts and reporting expectations.

3

Risk

Assess data, methodological, transition, physical and reputational GHG risks.

4

Review

Use controls, reconciliation, uncertainty evaluation and management review before reporting.

© 2026 Tertiary Infotech Academy Pte Ltd

K6 · K7 · A4 · A8 · LO2

Reporting requirements and conformance

1

Applicable obligations

Laws, schemes, contracts and lender or customer terms can create different requirements.

2

Reporting criteria

The inventory must name the standard, programme rules and reporting period used.

3

Mandatory disclosures

Boundaries, categories, methods, exclusions, uncertainty and changes need controlled evidence.

4

Approval

Management accepts responsibility for the GHG statement before external use.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Maintain an obligation register
- Translate each requirement into evidence
- Track submission and approval dates
- Control public claim versions

Avoid the trap

- Voluntary does not mean consequence-free
- Different users may need different views
- Legal applicability is context-specific
- Do not imply verification before it occurs

© 2026 Tertiary Infotech Academy Pte Ltd

K6 · K7 · A4 · A8 · LO2

Inventory data governance and internal control

1

Ownership

Every source has an accountable owner and competent preparer.

2

Method control

Factors, GWP values, calculations and estimates are versioned and approved.

3

Evidence control

Source records are complete, retrievable, protected and retained.

4

Review control

Independent checks, reconciliations and issue closure support a reliable statement.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Use a source-level RACI
- Lock formulas and controlled inputs
- Log changes and approvals
- Review high-risk sources first

Avoid the trap

- Sustainability should not own all data
- A spreadsheet can hide silent changes
- Review is more than proofreading
- One approval cannot compensate for weak evidence

© 2026 Tertiary Infotech Academy Pte Ltd

K6 · K7 · A4 · A8 · LO2

Reviewing a GHG inventory report

1

Boundary test

Confirm included entities, categories and significant exclusions match documented decisions.

2

Calculation test

Reperform samples from source record through unit conversion, factor and total.

3

Disclosure test

Check claims, estimates, limitations and changes are described accurately.

4

Trend test

Challenge whether structural change, method change or data gaps explain apparent performance.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Trace both top-down and bottom-up
- Sample anomalies and high magnitude
- Resolve contradictions
- Retest every correction

Avoid the trap

- A balanced total can contain offsetting errors
- A factor source without vintage is weak
- Narrative claims require the same control as numbers
- Verification readiness is not verification

© 2026 Tertiary Infotech Academy Pte Ltd

K6 · K7 · A4 · A8 · LO2

GHG risk landscape

1

Inventory risk

Incomplete sources, wrong units, weak factors, estimates or uncontrolled formulas distort results.

2

Transition risk

Policy, carbon cost, technology and market expectations can affect competitiveness.

3

Physical risk

Acute and chronic hazards can disrupt operations, data and value-chain performance.

4

Claims risk

Ambiguous boundaries or unsupported neutrality and reduction statements damage trust.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Write cause-event-consequence statements
- Assess current-control effectiveness
- Use thresholds and early indicators
- Link material risks to management review

Avoid the trap

- Risk is not only the reported number
- High uncertainty can be decision material
- Physical and transition risks can interact
- Greenwashing risk is a governance issue

© 2026 Tertiary Infotech Academy Pte Ltd

K6 · K7 · A4 · A8 · LO2

Risk-responsive GHG management plan

1

Objectives

Connect inventory reliability and reduction outcomes to strategy and obligations.

2

Controls

Prevent, detect and correct data, method, performance and claim failures.

3

Monitoring

Use leading and lagging indicators with thresholds and escalation.

4

Learning

Review changes, incidents, findings and stakeholder challenges to improve the system.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Prioritise by consequence and control weakness
- Assign decision rights
- Test response scenarios
- Report residual risk transparently

Avoid the trap

- A register without action is not management
- Too many KPIs obscure warning signals
- Risk acceptance needs authority
- Plans must evolve with business change

© 2026 Tertiary Infotech Academy Pte Ltd

Activities · Topic 2

Activity 4

- Review a GHG Inventory and Resolve Quality Findings
- K5 · K6 · A4 · LO2
- 75 minutes

Activity 5

- Build a GHG Governance, Compliance and Risk Plan
- K6 · K7 · A4 · A8 · LO2
- 60 minutes

Outputs

- Detailed procedure in LG
- Workbook and PDF pack
- Evidence-based acceptance test

© 2026 Tertiary Infotech Academy Pte Ltd

K5 · K6 · A4 · LO2 · 75 minutes

Review a GHG Inventory and Resolve Quality Findings

ACTIVITY 4

Learners act as an independent internal review team and test EcoForge's draft report, calculation workbook and control evidence.

You'll build

A review plan, evidence log, finding register, corrected disclosure set and management approval brief.

Tools: Draft inventory report · evidence pack · review workbook · finding rubric

© 2026 Tertiary Infotech Academy Pte Ltd

Activity 4 · Process

1. Set review criteria
2. Trace totals to evidence
3. Reperform calculations
4. Test disclosures
5. Resolve findings

© 2026 Tertiary Infotech Academy Pte Ltd

ACTIVITY 4 · ACCEPTANCE CHECK

Review a GHG Inventory and Resolve Quality Findings

✅ Test it

All seeded defects are detected; each finding is evidence-based; corrections are rechecked; the final brief distinguishes resolved issues from residual limitations.

© 2026 Tertiary Infotech Academy Pte Ltd

K6 · K7 · A4 · A8 · LO2 · 60 minutes

Build a GHG Governance, Compliance and Risk Plan

ACTIVITY 5

Learners translate EcoForge's reporting obligations and climate exposures into a practical governance and risk-management plan.

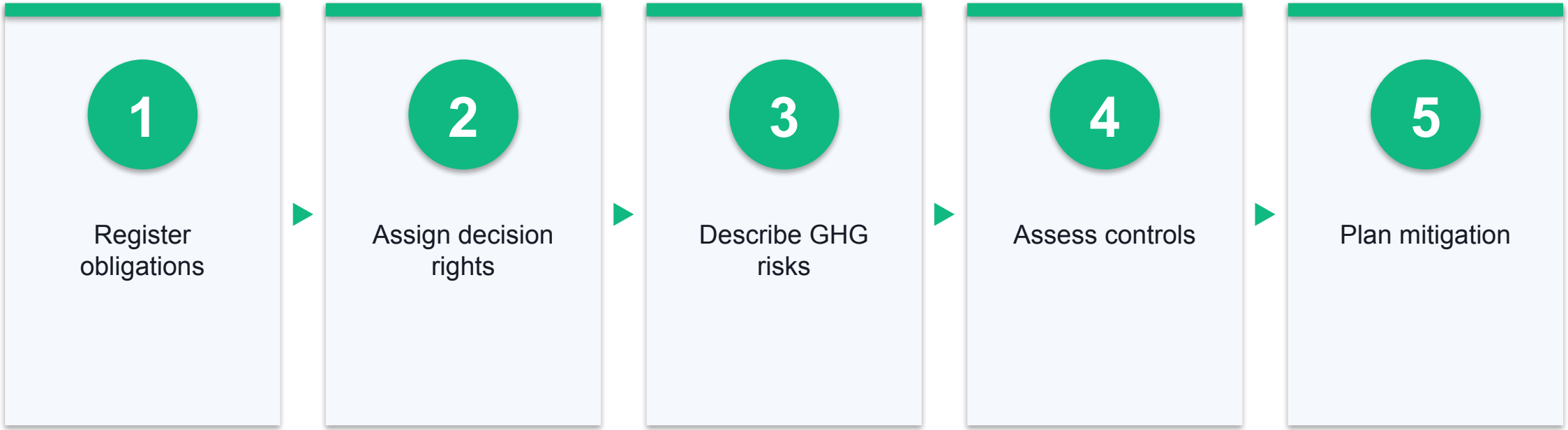
You'll build

A compliance register, RACI, GHG risk register, control plan, monitoring dashboard and escalation protocol.

Tools: Risk workbook · obligation cards · governance case pack · dashboard template

© 2026 Tertiary Infotech Academy Pte Ltd

Activity 5 · Process



© 2026 Tertiary Infotech Academy Pte Ltd

ACTIVITY 5 · ACCEPTANCE CHECK

Build a GHG Governance, Compliance and Risk Plan

✅ Test it

Every applicable obligation has an owner and evidence; high risks have proportionate controls and indicators; accountability and escalation are unambiguous.

© 2026 Tertiary Infotech Academy Pte Ltd

Recap · Regulatory-Compliant and Risk-Responsive GHG Management

1

Explain

Assign authority, competence, control ownership and escalation for inventory quality.

2

Apply

Evaluate inventory evidence, controls, conformance and report completeness before management approval or verification.

3

Demonstrate

Design roles, controls, compliance monitoring and mitigation actions for material GHG risks.

4

Evidence

Retain the completed activity artefacts and review feedback.

© 2026 Tertiary Infotech Academy Pte Ltd

DAY 2

Reduction Strategy → Carbon Strategy

03

TOPIC 03

Operationalising Emissions Reduction Across Value Chains

Boundaries · source data · baselines · lifecycle and supplier action · renewable energy

© 2026 Tertiary Infotech Academy Pte Ltd

K3 · K4 · A1 · A6 · LO3

Topic 03 · Core Model

1

Inventory-to-action

Use hotspot and trend evidence to direct reduction effort where it matters.

2

Hierarchy

Avoid and reduce first; substitute, recover and neutralise only where appropriate.

3

Value chain

Engage suppliers and customers using category-relevant data and decision levers.

4

Energy strategy

Compare efficiency, electrification and renewable-energy options on cost and integrity.

© 2026 Tertiary Infotech Academy Pte Ltd

K3 · K4 · A1 · A6 · LO3

From inventory to reduction hierarchy

1

Avoid

Remove unnecessary activity, demand or material before optimising it.

2

Reduce

Improve efficiency, yield, maintenance, logistics and process control.

3

Substitute

Change fuel, material, technology, product design or service model.

4

Neutralise residuals

Consider high-integrity removals or offsets only for clearly defined residual emissions.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Start with hotspots and influence
- Quantify expected abatement
- Check rebound and burden shift
- Separate forecast from realised result

Avoid the trap

- Largest is not always most controllable
- Low-cost action can have low impact
- Avoided emissions differ from inventory reductions
- Offsetting is not operational reduction

© 2026 Tertiary Infotech Academy Pte Ltd

K3 · K4 · A1 · A6 · LO3

Lifecycle and value-chain thinking

1

Upstream

Materials, capital goods, fuel supply, inbound logistics and supplier operations.

2

Operations

Energy, process emissions, refrigerants, waste and owned transport.

3

Downstream

Distribution, product use, maintenance, end of life and customer behaviour.

4

Interfaces

Design and procurement choices can shift emissions between stages and organisations.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Map physical flows before calculating
- Find decision levers at interfaces
- Use functional comparisons
- Check end-of-life assumptions

Avoid the trap

- Spend is not a lifecycle model
- Recycling claims need yield and allocation
- Efficiency can trigger rebound
- Do not double count supplier and product action

© 2026 Tertiary Infotech Academy Pte Ltd

K3 · K4 · A1 · A6 · LO3

Supplier data and engagement ladder

1

Estimate

Screen with spend, mass or industry-average factors while labelling limitations.

2

Request

Ask for activity, facility or product data using a clear specification.

3

Collaborate

Improve methods, capability and reduction plans with strategic suppliers.

4

Contract and verify

Embed criteria, evidence, improvement and assurance expectations into sourcing.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Segment by emissions and influence
- Standardise data requests
- Reward data improvement
- Track both coverage and reduction

Avoid the trap

- A supplier claim may use a different boundary
- Primary data can still be poor
- Questionnaires create burden
- Changing supplier may export risk rather than reduce it

© 2026 Tertiary Infotech Academy Pte Ltd

K3 · K4 · A1 · A6 · LO3

Reduction baselines, targets and monitoring

1

Baseline

State period, boundary, method, quality and structural-change treatment.

2

Target

Define absolute or intensity metric, coverage, ambition, milestones and responsibility.

3

Pathway

Model business-as-usual and initiative effects without hiding growth assumptions.

4

Monitoring

Track implementation, realised activity change, emission factors and residual gaps.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Use annual and interim milestones
- Reconcile projects to inventory
- Separate activity and factor effects
- Reforecast when assumptions change

Avoid the trap

- Intensity can fall while absolute emissions rise
- Targets do not prove achievement
- Forecast abatement may overlap
- Grid decarbonisation is not an internal project

© 2026 Tertiary Infotech Academy Pte Ltd

K3 · K4 · A1 · A6 · LO3

Renewable energy and economic integrity

1

Efficiency first

Reduce demand to lower cost, capacity and exposure.

2

Onsite generation

Model technical yield, self-consumption, export, maintenance and asset life.

3

Electrification

Evaluate process suitability, load, grid trajectory, resilience and capital needs.

4

Procurement

Assess contract, certificate geography, vintage, exclusivity, retirement and claim rules.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Compare NPV and cost per tCO2e
- Run tariff and yield sensitivities
- State inventory accounting treatment
- Stage investment with decision gates

Avoid the trap

- A certificate does not reduce physical onsite use
- Market-based claims require controls
- Cheap credits can distract from efficiency
- Technology choice must preserve safety and quality

© 2026 Tertiary Infotech Academy Pte Ltd

Activities · Topic 3

Activity 6

- Diagnose Hotspots and Set a Reduction Baseline
- K3 · A1 · LO3
- 50 minutes

Activity 7

- Design a Supplier and Product Reduction Roadmap
- K3 · K9 · A1 · LO3
- 60 minutes

Activity 8

- Develop a Renewable-Energy Business Case
- K4 · A6 · LO3
- 55 minutes

© 2026 Tertiary Infotech Academy Pte Ltd

K3 · A1 · LO3 · 50 minutes

Diagnose Hotspots and Set a Reduction Baseline

ACTIVITY 6

Learners convert EcoForge's inventory into a prioritised hotspot and abatement-opportunity view without confusing spend, emissions and influence.

You'll build

A hotspot dashboard, lifecycle map, baseline statement and opportunity shortlist.

Tools: Inventory workbook · lifecycle map · hotspot scoring tool

© 2026 Tertiary Infotech Academy Pte Ltd

Activity 6 · Process

1. Rank emission hotspots
2. Map lifecycle stages
3. Test burden shifts
4. Define the baseline
5. Prioritise reduction levers

© 2026 Tertiary Infotech Academy Pte Ltd

ACTIVITY 6 · ACCEPTANCE CHECK

Diagnose Hotspots and Set a Reduction Baseline

✅ Test it

Priorities reflect magnitude, influence and uncertainty; baseline and intensity denominator are explicit; no recommendation creates an undisclosed lifecycle burden shift.

© 2026 Tertiary Infotech Academy Pte Ltd

K3 · K9 · A1 · LO3 · 60 minutes

Design a Supplier and Product Reduction Roadmap

ACTIVITY 7

Learners build a phased roadmap that improves EcoForge's supplier data and reduces embodied, logistics and product-use emissions.

You'll build

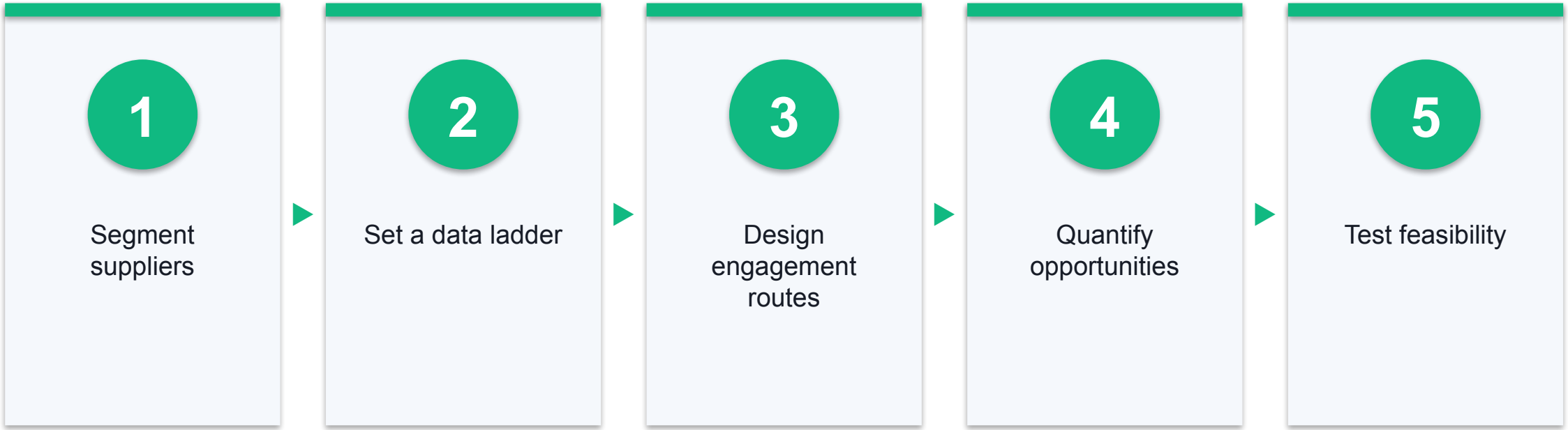
A 24-month reduction roadmap, supplier data ladder, initiative cards, KPI set and implementation governance.

Tools: Supplier portfolio workbook · initiative design cards · roadmap canvas

© 2026 Tertiary Infotech Academy Pte Ltd

K3 · K9 · A1 · LO3 · HIGH-LEVEL WORKFLOW

Activity 7 · Process



© 2026 Tertiary Infotech Academy Pte Ltd

ACTIVITY 7 · ACCEPTANCE CHECK

Design a Supplier and Product Reduction Roadmap

✅ Test it

The roadmap contains quantified, owned and time-bound actions; supplier data improves progressively; product actions address lifecycle trade-offs and claim boundaries.

© 2026 Tertiary Infotech Academy Pte Ltd

K4 · A6 · LO3 · 55 minutes

Develop a Renewable-Energy Business Case

ACTIVITY 8

Learners advise EcoForge on a portfolio of onsite solar, efficiency, electrification and renewable-electricity procurement options.

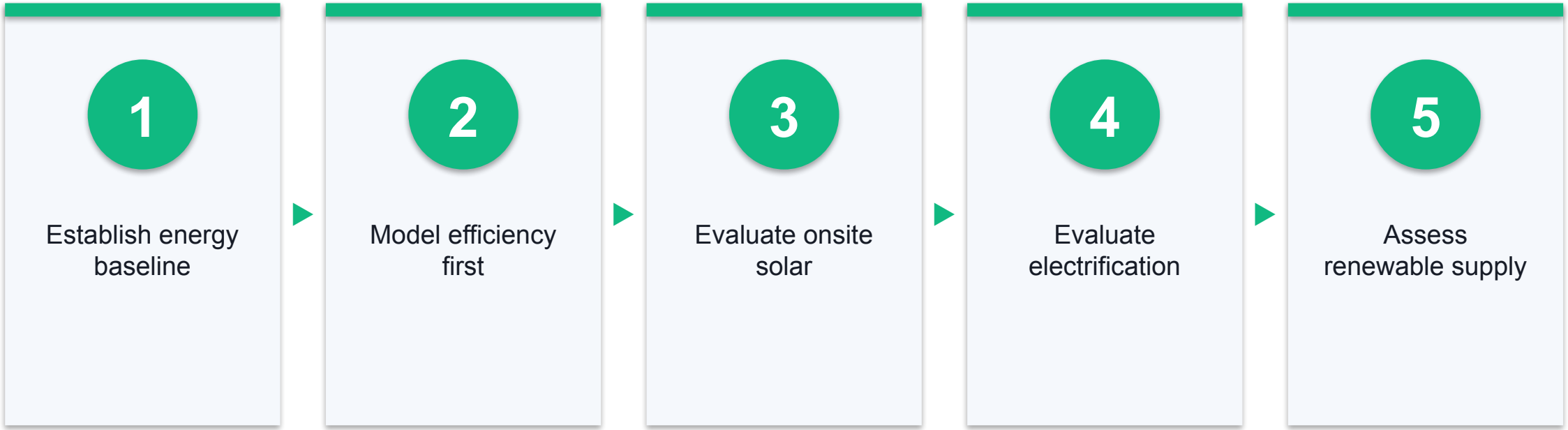
You'll build

A scenario model, marginal-abatement view, option scorecard and executive recommendation.

Tools: Energy profile workbook · tariff and option facts · financial model

© 2026 Tertiary Infotech Academy Pte Ltd

Activity 8 · Process



© 2026 Tertiary Infotech Academy Pte Ltd

ACTIVITY 8 · ACCEPTANCE CHECK

Develop a Renewable-Energy Business Case

✅ Test it

The recommendation uses consistent assumptions, separates demand and supply levers, quantifies economics and emissions, and states certificate and claim safeguards.

© 2026 Tertiary Infotech Academy Pte Ltd

Recap · Operationalising Emissions Reduction Across Value Chains

1

Explain

Use hotspot and trend evidence to direct reduction effort where it matters.

2

Apply

Use inventory, lifecycle and business data to identify hotspots, set a defensible baseline and frame reduction opportunities.

3

Demonstrate

Develop quantified value-chain actions, supplier engagement, product levers and monitoring milestones.

4

Evidence

Retain the completed activity artefacts and review feedback.

© 2026 Tertiary Infotech Academy Pte Ltd

04

TOPIC 04

Designing Viable Carbon Offset and Strategy Plans

Residual emissions · project quality · offset economics · claims · carbon-credit optimisation

© 2026 Tertiary Infotech Academy Pte Ltd

K8 · K9 · A2 · A5 · LO4

Topic 04 · Core Model

1

Residual first

Offsets address clearly defined residual emissions after credible internal reductions.

2

Quality

Evaluate additionality, permanence, leakage, quantification, verification and double counting.

3

Economics

Compare price, transaction cost, delivery risk, durability and strategic co-benefits.

4

Claims

Use precise boundaries, retirement evidence and transparent language that avoids overstatement.

© 2026 Tertiary Infotech Academy Pte Ltd

K8 · K9 · A2 · A5 · LO4

Residual emissions and the role of offsets

1

Gross inventory

Report organisational emissions and removals before external credits.

2

Reduction pathway

Show internal and value-chain actions against a declared baseline.

3

Residual emissions

Define what remains after feasible reduction within the target boundary.

4

Credit use

Disclose quantity, project, vintage, ownership, retirement and claim separately.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Set an offset eligibility policy
- Forecast residuals conservatively
- Procure against a clear use case
- Retire credits before claiming use

Avoid the trap

- Credits are not removals by the organisation
- Future delivery may fail
- Neutrality language can overstate scope
- Do not hide gross performance

© 2026 Tertiary Infotech Academy Pte Ltd

Carbon project integrity

1

Additionality

The claimed outcome would not occur without the project incentive under the defined test.

2

Baseline and quantification

A credible counterfactual, monitoring method and uncertainty treatment support the tonnes.

3

Permanence and leakage

Durability, reversal response and displaced emissions are addressed.

4

No double counting

Issuance, ownership, transfer, host claims and retirement prevent duplicate use.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Review methodology and project documents
- Check independent validation and verification
- Inspect registry status
- Assess safeguards and grievance routes

Avoid the trap

- Registry listing is not automatic quality
- Co-benefits do not repair weak carbon accounting
- Avoidance and removal have different durability
- Buffer pools do not eliminate reversal risk

© 2026 Tertiary Infotech Academy Pte Ltd

K8 · K9 · A2 · A5 · LO4

Offset economics and procurement risk

1

Unit price

Quoted cost per credit is only the starting point.

2

Total cost

Include due diligence, legal, registry, retirement, monitoring and internal governance.

3

Delivery risk

Forward projects face issuance, timing, volume, policy and counterparty uncertainty.

4

Replacement exposure

Low-quality or failed credits may require replacement at a higher future price.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Model risk-adjusted cost
- Use portfolio limits
- Set conditions precedent
- Retain retirement and claim evidence

Avoid the trap

- Lowest price can carry highest risk
- Co-benefits need evidence
- Vintage must match intended use
- A long contract can lock in weak criteria

© 2026 Tertiary Infotech Academy Pte Ltd

K8 · K9 · A2 · A5 · LO4

Carbon-credit portfolio optimisation

1

Purpose

Match credit type and timing to residual emissions and claim need.

2

Quality floor

Define non-negotiable methodology, verification, safeguards and registry conditions.

3

Diversification

Manage project, geography, technology, delivery and counterparty concentration.

4

Governance

Set approval, custody, retirement, accounting, disclosure and review responsibilities.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Use scoring plus hard exclusions
- Stage procurement over time
- Reserve for delivery failure
- Review criteria as science and policy evolve

Avoid the trap

- Diversification cannot legitimise weak credits
- Removal scarcity affects price
- Portfolio optimisation is not emissions optimisation
- Claims need boundary-specific wording

© 2026 Tertiary Infotech Academy Pte Ltd

K8 · K9 · A2 · A5 · LO4

Integrated organisational carbon strategy

1

Inventory foundation

A governed, repeatable inventory establishes what is measured and managed.

2

Target and pathway

Business growth, initiatives, dependencies and residuals are reconciled transparently.

3

Capital and risk

Investment gates balance abatement, economics, resilience, operations and data confidence.

4

Disclosure and learning

A dashboard, annual review and claim controls keep strategy credible as conditions change.

© 2026 Tertiary Infotech Academy Pte Ltd

Decision Lens

Apply in the organisation

- Connect every initiative to an inventory source
- Assign accountable owners and budgets
- Stress-test key assumptions
- Report gross, reduced and offset tonnes separately

Avoid the trap

- A strategy is not a list of projects
- Targets require governance and recalculation
- Do not combine avoided and inventory emissions
- Update claims when evidence changes

© 2026 Tertiary Infotech Academy Pte Ltd

Activities · Topic 4

Activity 9

- Evaluate Carbon Offset Projects and Economics
- K8 · K9 · A2 · LO4
- 50 minutes

Activity 10

- Integrate the Organisational Carbon Strategy
- K2 · K7 · K8 · K9 · A2 · A5 · LO4
- 30 minutes

Outputs

- Detailed procedure in LG
- Workbook and PDF pack
- Evidence-based acceptance test

© 2026 Tertiary Infotech Academy Pte Ltd

K8 · K9 · A2 · LO4 · 50 minutes

Evaluate Carbon Offset Projects and Economics

ACTIVITY 9

Learners compare three fictional project proposals and decide whether any are suitable for EcoForge's residual-emissions strategy.

You'll build

An offset due-diligence register, quality scorecard, total-cost model and procurement recommendation.

Tools: Project dossiers · quality rubric · economics workbook · claims checklist

© 2026 Tertiary Infotech Academy Pte Ltd

Activity 9 · Process

1. Define intended use
2. Review additionality
3. Assess permanence and leakage
4. Verify registry controls
5. Score safeguards and value

© 2026 Tertiary Infotech Academy Pte Ltd

ACTIVITY 9 · ACCEPTANCE CHECK

Evaluate Carbon Offset Projects and Economics

✅ Test it

The decision is evidence-based, includes quality and financial risk, requires retirement proof, and does not use credits to obscure gross organisational emissions.

© 2026 Tertiary Infotech Academy Pte Ltd

K2 · K7 · K8 · K9 · A2 · A5 · LO4 · 30 minutes

Integrate the Organisational Carbon Strategy

ACTIVITY 10

Learners complete the EcoForge capstone by presenting a 2030 carbon strategy to an executive committee under cost, growth and credibility constraints.

You'll build

A 2030 target pathway, initiative portfolio, residual-emissions forecast, offset policy, carbon-credit plan and board dashboard.

Tools: Capstone workbook · initiative portfolio · decision memo template · board role cards

© 2026 Tertiary Infotech Academy Pte Ltd

Activity 10 · Process

1. Consolidate the evidence
2. Define target architecture
3. Model the pathways
4. Sequence reduction actions
5. Govern residual emissions

© 2026 Tertiary Infotech Academy Pte Ltd

ACTIVITY 10 · ACCEPTANCE CHECK

Integrate the Organisational Carbon Strategy

✅ Test it

The strategy reconciles baseline, pathway and residuals; reduction precedes offsetting; financial and delivery risks are explicit; targets, credits and claims have accountable governance.

© 2026 Tertiary Infotech Academy Pte Ltd

Recap · Designing Viable Carbon Offset and Strategy Plans

1

Explain

Offsets address clearly defined residual emissions after credible internal reductions.

2

Apply

Assess offset quality, delivery and financial implications before procurement.

3

Demonstrate

Combine inventory governance, reduction pathways, renewable energy and high-integrity offsets into an executable strategy.

4

Evidence

Retain the completed activity artefacts and review feedback.

© 2026 Tertiary Infotech Academy Pte Ltd

WRAP-UP

From Inventory Evidence to Carbon Strategy

© 2026 Tertiary Infotech Academy Pte Ltd

COURSE OUTCOMES

What You Achieved

1

LO1

Innovate organisational GHG accounting systems using current climate science, ISO 14064-1 principles and broader sustainability initiatives.

2

LO2

Develop risk-informed GHG management plans that meet relevant reporting and regulatory requirements and support mitigation decisions.

3

LO3

Develop operational emissions-reduction strategies across product lifecycles, value chains, supply chains and renewable-energy options.

4

LO4

Generate economically viable carbon-offset and carbon-credit strategies aligned with organisational targets and residual-emissions needs.

© 2026 Tertiary Infotech Academy Pte Ltd

END-TO-END PRACTICE

The Complete GHG Management Cycle

1. Set purpose and boundaries
2. Quantify and review
3. Assess risk and hotspots
4. Reduce and invest
5. Report, assure and improve

© 2026 Tertiary Infotech Academy Pte Ltd

https://lms-tms.tertiaryinfotech.com/

Courseware and Assessment on the LMS

1

Download

Get the learner slides PDF and Learner Guide.

2

Activities

Open the complete Activities folder and working files.

3

Assessment

Complete the WA and Case Study during the scheduled window.

4

Submit

Upload completed answers and confirm successful submission.

© 2026 Tertiary Infotech Academy Pte Ltd

FINAL REMINDER

Assessment

WA

- 9 questions
- K1-K9
- 1 hour · open book

Case Study

- 4 questions
- A1-A8
- 1 hour · open book

Complete

- Assessment attendance
- Submit on LMS
- Sign summary record

© 2026 Tertiary Infotech Academy Pte Ltd

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM digital attendance
2. Assessment digital attendance
3. Sit WA then Case Study
4. Submit answers on the LMS
5. Sign Assessment Summary Record

© 2026 Tertiary Infotech Academy Pte Ltd

Digital Attendance (Mandatory)

1

AM, PM and assessment

Complete every required SSG digital-attendance session.

2

Scan

Use the SSG QR code shown by the trainer or administrator.

3

Confirm

Wait for the successful submission acknowledgement.

4

Eligibility

At least 75% attendance is required for assessment and funding.

© 2026 Tertiary Infotech Academy Pte Ltd

THANK YOU

Account for what matters. Reduce what you can. Disclose what remains.

Use transparent evidence to turn an organisational inventory into credible climate decisions.

© 2026 Tertiary Infotech Academy Pte Ltd